2026-10-16 | VS&A Daily |

Before the hammer: pre-test analysis for a small finned rocket



Black Brant IX carrying Dynamo-2 from Wallops, July 2021. A real finned sounding rocket provides context; this is not a modal-test photograph or the worked-example vehicle. Credit: NASA/Allison Stancil. NASA image source [1].

Four accelerometers. A slender airframe. A deceptively easy plan.

Put the sensors along the body, tap sideways, and the bending peaks should appear. But what would that experiment tell you about fin twist, the other bending plane, or a torsional mode with almost no radial motion? A neat instrumentation drawing can leave the most important question unasked.

Today’s objective is to turn a pre-test finite element model into a defensible measurement and excitation plan. This expanded edition adds solved body and fin component FEMs, mode-shape plots, and a same-channel-count comparison of equal, irregular and model-selected spacing.

Allow about 15 minutes. These are illustrative component studies, not a coupled flight-vehicle model or measured rocket data. We build on the October 9 spatial-aliasing lesson and ask a sharper question: should the best sensor layout look evenly spaced, symmetric, both, or neither?

Define the experiment before choosing the sensors

First decide what the correlated model must predict: global loads, fin-root stress, aeroelastic behavior, or control-sensitive bending. Set the upper frequency and target list accordingly. Include both lateral bending planes and close pairs, axial and body-torsional modes where relevant, and local fin bending, twist and root coupling. A mode with little total effective mass can still matter greatly to one fin attachment.

Freeze the tested configuration: motor or inert mass simulator, payload, fasteners, joint preload and wiring. Check total mass, center of gravity and inertia against the FEM; verify connectivity, units, mesh convergence and the six free rigid-body modes. Flight and test configurations need separate, traceable definitions.

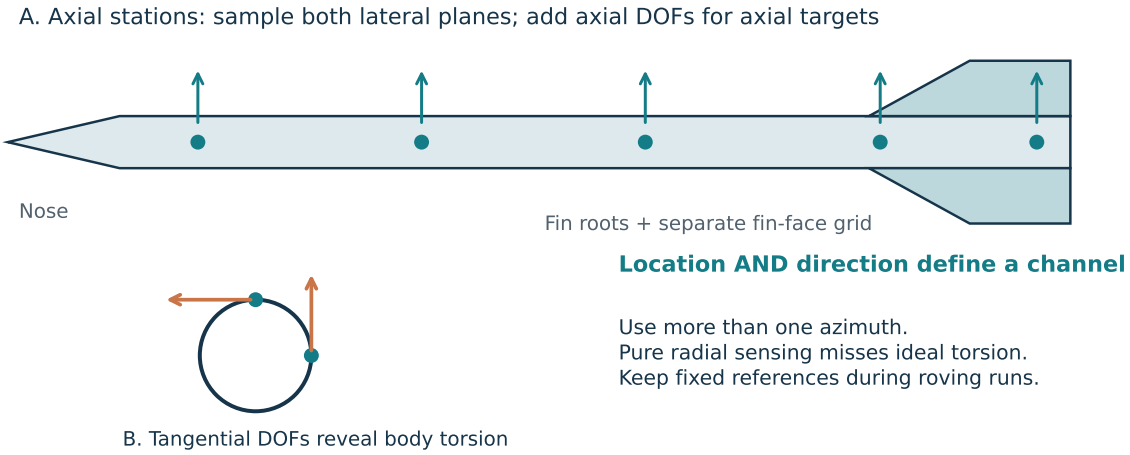


Figure 1. Original candidate-layout concept, not a prescribed channel count. Arrows indicate selected sensing directions; the second bending plane is out of the side-view page. Tangential arrows in the cross-section distinguish torsion from radial motion.

For a nominal free-free survey, represent the suspension rather than assuming it vanishes. KARI’s engine campaign screened rigid-body frequencies against the first elastic mode: below 10% for components and below 20% for larger assemblies. Its complete-engine model included suspension effects when a softer setup was impractical [2]. Treat those ratios as documented screening precedents, not proof of negligible coupling.

Move supports in the model and vary their stiffness. A first-bending node is not a node of every target mode. Instrument supports sufficiently to recognize their motion. For a fixture-mounted test, model the fixture and measure across its interface. Match the experiment you can actually perform.

Begin with generous coverage of load paths, joints and predicted modal motion, then reduce the candidate set. NASA’s ISS Node study used that engineering-judgment-first approach before algorithmic selection [3]. On a small rocket, protect fin and joint coverage when a global metric tries to discard it.

Make the FEM take the proposed test

Export the target eigenvectors at accessible sensor locations. Project each translation onto the installed sensing axis, including interpolation and coordinate transformations. With full-model modes Φ and measurement mapping S, the shapes the instrument can see are:

$$\Psi = S\Phi, \qquad a = \Psi\ddot{q} + \varepsilon$$

A triaxial package provides three channels, but not necessarily three useful directions for every target. Examine each column of Ψ for weak response, and compare columns with AutoMAC for shape ambiguity. Inspect singular values after stating the modal normalization and channel weighting; a small singular value flags a poorly distinguishable combination, even when individual ordinates look large.

Here a is measured acceleration, q contains modal amplitudes and ε is measurement noise. With noise covariance R, a useful information matrix is:

$$F = \Psi^T R^{-1} \Psi$$

F describes information about the chosen modal accelerations. Effective Independence selects coordinates to preserve independence of that modal basis. Full column rank requires at least as many independent rows as retained columns for this algebraic reconstruction problem. That is not a universal minimum sensor count for identifying separated resonances across frequencies or roving runs.

ECSS modal-survey guidance connects measurement directions, linear independence, AutoMAC and test predictions [4]. NASA’s ETM-3 motor survey paired Effective Independence selection with test-load simulations [5]. The transferable lesson is to evaluate the experiment, not just optimize a matrix.

Predict acceleration/force FRFs for feasible inputs, realistic damping ranges and the proposed array. A sideways hammer strike may excite bending well but hardly excite torsion; use tangential or offset forces and fin-normal inputs where needed. An impact hammer is efficient for reconnaissance. Shakers provide controlled sustained excitation and level sweeps, but their stingers, mass and load paths become part of the test.

Repeat the screening over plausible joint stiffness, support properties and sensor masses. Keep fixed references with visibility of the target families during roving measurements, and provide a backup reference where loss of one channel would break the dataset. An optimal nominal layout that fails under modest model error is not robust.

A solved fin model exposes the blind spot

Take a rectangular aluminum fin: 150 mm span, 100 mm chord, 2 mm thickness; E = 70 GPa, ν = 0.33 and density 2700 kg/m³. Clamp its root. Use 20 cubic-Hermite beam elements for out-of-plane bending and 20 linear torsion elements about mid-chord, both with consistent mass. The two fields are uncoupled: this is a strip model with rigid chordwise sections, not a plate or fin-root/body model.

The computed first bending and twist frequencies are 73.11 and 206.83 Hz. Doubling the mesh changes each by less than 0.02%; both agree with their corresponding beam/rod closed forms within 0.03%. That verifies this idealization's calculation, not its fidelity to a real fin.

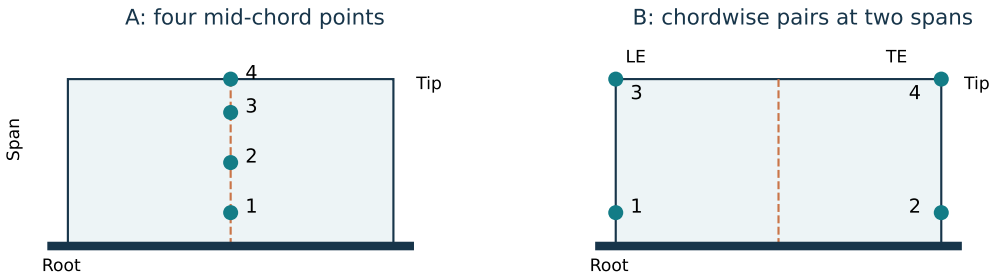


Figure 2. Original FE sampling diagram. A: span fractions 0.2, 0.5, 0.8, 1 at mid-chord. B: leading/trailing edges at span fractions 0.2 and 1. All four channels sense normal to the page; dashed line is the ideal twist axis.

Channel	A: bending	A: twist	B: bending	B: twist
1	0.064	0.000	0.064	0.309
2	0.340	0.000	0.064	-0.309
3	0.725	0.000	1.000	1.000
4	1.000	0.000	1.000	-1.000

Table 1. Computed dimensionless FE displacements, each mode normalized to its maximum fin-face motion. Twist displacement at chord offset e is e times the torsional rotation; its sign convention is arbitrary.

A has rank one: twist is invisible, so its twist AutoMAC is undefined. B has rank two. Its paired channels retain equal bending and opposite twist, giving:

$$\mathrm{MAC}(b,t)=\frac{|b^{T}t|^{2}}{(b^{T}b)(t^{T}t)}=0 \quad \text{for layout B}$$

Symmetric placement helped here. Real taper, joint flexibility and bend-twist coupling require shell/solid or validated beam models. Add adjacent-body/root channels, and apply an off-axis fin force: a mid-chord normal force cannot excite this ideal torsion mode.

Run a body FEM before drawing the station grid

Now isolate axial spacing. Represent one bending plane of a generic body by a uniform free-free Euler-Bernoulli beam: length L = 2 m, bending rigidity EI = 12,000 N·m² and line mass μ = 4 kg/m. These are assumed equivalent properties, not values fitted to a published rocket. Forty equal elements give 41 nodes and 82 displacement/rotation DOFs [11].

$$\mathbf{K}\boldsymbol{\phi}_r = \omega_r^2 \mathbf{M}\boldsymbol{\phi}_r, \qquad f_r = \frac{\omega_r}{2\pi}$$

Assemble standard cubic-Hermite stiffness and consistent mass matrices, impose no supports, and remove the two planar rigid-body modes from the elastic target set. The translational mass check returns 8.00 kg. Solve the first four elastic modes and retain only their transverse displacement rows for accelerometer selection.

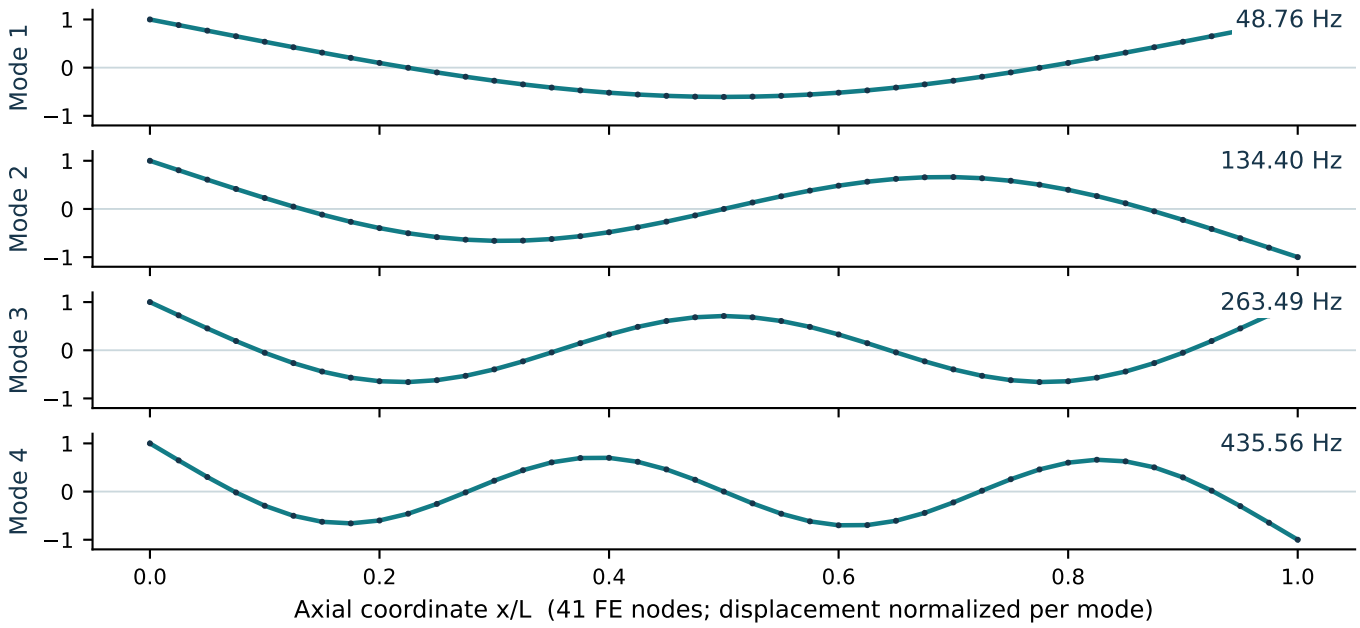


Figure 3. Original computed FE mode shapes. Dots are the 41 nodes, not proposed accelerometers. Each mode has unit maximum displacement; amplitude is arbitrary. Frequencies are 48.76, 134.40, 263.49 and 435.56 Hz.

Verification matters before optimization. Refining to 80 elements changes these frequencies by less than 0.002%. Agreement with the free-free beam solution, whose elastic roots satisfy cosh(β) cos(β) = 1, is also within 0.002%. These checks test the numerical model; they do not validate beam theory for shell ovalization, local fittings or fin coupling.

This example cannot select the complete rocket array: it excludes the other bending plane, axial/torsional motion, fins and suspension. It demonstrates one reproducible step to repeat using the corresponding target families of a verified three-dimensional model.

Equal, irregular, or selected from the modes?

Hold the budget at six transverse channels for those four body modes. Start EfI from 21 candidates at x/L = 0, 0.05, ..., 1. With equal independent channel noise, compute the row leverage below, remove the smallest contributor, and recompute until six remain [8,9]. This greedy result is not a guaranteed global optimum.

$$h_i = \left[\boldsymbol{\Psi}(\boldsymbol{\Psi}^T\boldsymbol{\Psi})^{-1}\boldsymbol{\Psi}^T\right]_{ii}$$

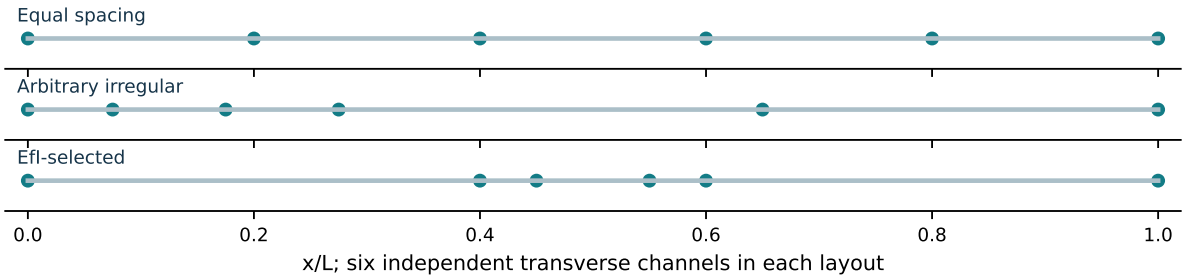


Figure 4. Original layouts evaluated with the same FEM. The arbitrary irregular case is intentionally just an unscreened alternative, not representative of all irregular arrays.

Layout / x/L coordinates	Max off-diag. MAC	Condition κ_2
Equal spacing 0, 0.2, 0.4, 0.6, 0.8, 1	0.315	1.89
Arbitrary irregular 0, 0.075, 0.175, 0.275, 0.65, 1	0.640	3.25
EfI-selected 0, 0.4, 0.45, 0.55, 0.6, 1	0.132	1.46

Table 2. Computed geometric distinguishability. For κ_2, normalize each sampled column to unit Euclidean norm, then divide the largest singular value by the smallest. Lower values are preferable here; these are not acceptance thresholds or absolute signal-to-noise estimates.

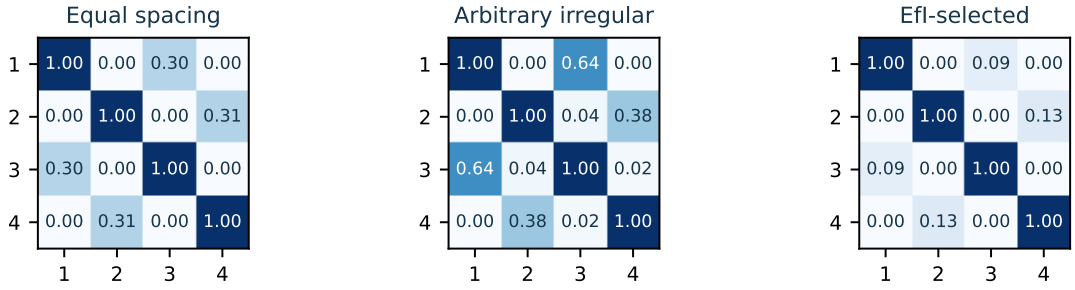


Figure 5. AutoMAC of the four sampled body modes; axes are mode numbers 1-4. The symmetric arrays preserve near-zero cross terms between opposite-parity modes. Same-parity modes still need sufficient axial coverage.

EfI improves this example while retaining mirror symmetry and unequal spacing. Unequal spacing alone is not the reason it performs better: the arbitrary irregular set performs worse.

Use rules of thumb as questions to test

Spatial spacing should resolve the shortest target deformation pattern, not follow a fixed centimetre rule. More than two samples per sinusoidal wavelength avoids the ideal Nyquist ambiguity, but barely meeting that limit does not yield a reliable finite-structure mode shape. Check the actual sampled vectors, including chordwise and circumferential coverage. Large antinode response helps signal quality; it does not ensure independence.

As one sensitivity check, add 0.4 kg at x/L = 0.75 to the body FEM. Without reselecting sensors, the equal, irregular and EfI layouts give maximum off-diagonal MACs of 0.305, 0.612 and 0.133. The ranking survives this perturbation; broader joint, support and mass variations still need evaluation.

Sensor mass should be judged against local modal inertia, not just total rocket mass. For a small added point mass m_s, negligible added stiffness and an otherwise unchanged mode, first-order eigenvalue perturbation gives:

$$\frac{\Delta f_r}{f_r} \approx -\frac{m_s \phi_r(x_s)^2}{2m_r}$$

Here m_r is generalized modal mass using the same normalization as the local translational ordinate φ_r. For a tip-normalized fin mode with m_r = 20 g, a 1 g sensor at unit ordinate predicts about a 2.5% frequency decrease. This illustrative estimate excludes adhesive, cable stiffness and mode-shape changes; verify by mass perturbation or an alternative measurement. A small fraction of vehicle mass can be a large fraction of fin modal mass.

Frequency resolution follows record duration, not sampling rate alone: $\Delta f = 1/T$. For illustrative modes near 80 Hz separated by 1 Hz and 0.5% damping, the light-damping bandwidth is about 0.8 Hz. An 8 s record gives 0.125 Hz lines, a reasonable starting grid to test, not guaranteed separation. Also verify adequate decay, leakage treatment, anti-alias filtering and usable force spectrum.

A real sounding-rocket case makes the distinction concrete. Faraday Rocketry’s published test used four uniaxial accelerometers and compared two bending frequencies with its FEM. Yet the account explicitly reports poor mode-shape confidence and limited spatial resolution [6]. Frequency agreement alone cannot establish fin-twist observability or readiness for flutter prediction.

The same account describes low suspension frequencies as desirable but labels measured modes near 18-20 Hz as suspension motion, versus first bending near 29 Hz [6]. That discrepancy deserves investigation, not adoption as best practice. The case is valuable evidence of practical limitations; its broad validation language and generic frequency-error tolerance are not universal acceptance criteria.

Keep useful symmetry; challenge blind spots

Equal spacing and symmetry are different choices. Our EfI result is nonuniform and symmetric. The fin edge pairs are symmetric too. For a symmetric structure, an even mode has matching motion at mirrored coordinates; an odd mode has opposite motion. Keeping both signed channels lets sums and differences separate the two. Averaging them first can erase the odd mode. Using only a symmetry plane can miss it entirely.

The literature agrees that geometry alone is not the selection rule. Kammer and Yao select sensors to preserve target-mode independence [8]. NASA implemented that approach directly in a Nastran normal-modes workflow [9]. Li and colleagues even obtained a uniform, symmetric arrangement for their particular suspension bridge [10]. That civil-structure example refutes a universal ban on symmetry; it does not prescribe a rocket layout.

For the rocket, begin with broad axial coverage and paired directions where they distinguish bending, twist or near-degenerate families. Then move or add coordinates where the sampled FEM shows weak participation or ambiguity. Do not randomize a sound layout merely to make it asymmetric. Inspect higher modes outside the selected set before accepting the central clustering of this example’s EfI array.

Excitation has the same geometry problem. In the body FEM, a center transverse force has zero participation in the antisymmetric second and fourth elastic modes. A transverse end force participates in all four. An improved response array cannot rescue a mode that the applied force does not excite. Real tests need accessible, load-limited inputs and predicted FRFs, not simply the largest modal ordinate.

Before the formal survey, check ambient noise, channel polarity and axes, mounting, cable restraint, hammer double hits and force bandwidth. Repeat FRFs, vary force level, and check reciprocity where the structure is linear, passive and reciprocal. Coherence is diagnostic, not a standalone pass/fail number; low values near antiresonances need context. KARI’s campaign explicitly used ambient, rigid-body, reciprocity and drive-point checks [2].

Correlate frequencies, shapes and FRFs on measured coordinates; for close pairs, compare subspaces. Expanded coordinates remain FEM inference. Update physically justified parameters and check withheld responses.

Three takeaways

1. Judge the sampled modes, not how tidy the layout looks.
2. Nonuniform spacing can help while useful symmetry is retained.
3. Verify the FEM and screen both sensor and force participation.

Reflection. The symmetric EfI array performs best here. Should you stagger it?
Answer. Only if model, uncertainty or checkout evidence improves the case. Asymmetry alone is not an objective.

Sources, model assumptions and reproducibility

[1] NASA SVS, Australia Sounding Rocket Campaign Press Kit, released June 7, 2022; Dynamo-2 photograph WFF-2021-045-002, July 2021, Allison Stancil. Photo and credit.

[2] J. Park et al., A Study on the Vibration Modal Testing and Analytical Model Updating of Liquid Rocket Engine of KSLV-II, EUCASS 2019, §§2.2-2.3. Original paper.

[3] M. L. Tinker, Accelerometer Placement for the International Space Station Node Modal Test, AIAA-98-2078, 1998, pp. 1-2. NASA copy.

[4] ECSS-E-ST-32-11C, Modal survey assessment, July 31, 2008, §§5.3 and 6.3. Space-program requirements subject to project applicability and tailoring. Standard.

[5] D. Nielsen et al., Modal Survey of ETM-3, A 5-Segment Derivative of the Space Shuttle Solid Rocket Booster, 2005. Methodological analogue, not a small finned rocket. NASA report.

[6] Dewesoft / Faraday Rocketry, Sounding Rocket Modal Testing for Structural Validation, August 28, 2026; setup, results and limitations. Participant case account.

[7] NASA-STD-5002B, March 10, 2026, §§1.2.1 and 4.9.1. Its pretest mass-orthogonality checks are useful context, but its scope explicitly excludes launch vehicles and sounding-rocket payloads. Do not import its criteria as this rocket's requirements. Standard.

[8] D. C. Kammer and L. Yao, Enhancement of On-Orbit Modal Identification of Large Space Structures Through Sensor Placement, Journal of Sound and Vibration 171(1), 119-139, March 17, 1994. Original EfI study; publisher abstract consulted. Paper.

[9] J. Peck and I. Torres, A DMAP Program for the Selection of Accelerometer Locations in MSC/NASTRAN, 2004. EfI implementation and five-segment rocket-booster pretest example. NASA paper.

[10] B. Li, D. Li, X. Zhao and J. Ou, Optimal sensor placement in health monitoring of suspension bridge, Science China Technological Sciences 55, 2039-2047, published April 11, 2012. Publisher abstract reports a uniform, symmetric result for that bridge. Paper.

[11] TU Delft, Computational Modelling: Euler-Bernoulli beam elements, undated online teaching text, accessed October 16, 2026. Hermite interpolation and element stiffness derivation. Derivation.

Calculation record

The PDF includes the Python calculation script and JSON results as embedded attachments for readers that support PDF attachments. NumPy/SciPy solve real symmetric generalized eigenproblems. Fin bending uses $I = ct^3/12$; torsion uses $G = E/[2(1+\nu)]$, $J \approx ct^3(1-0.63t/c)/3$ and polar area moment $I_p = ct(c^2+t^2)/12$. The torsion model neglects warping restraint. No sensor mass, joints or support stiffness is included in the baseline FEMs; the body mass perturbation is separate.

Expanded October 16, 2026. Original figures and tables show computed illustrative models, not measured instrumentation or telemetry. These component studies do not represent a coupled finned-rocket model. Sources are public; no proprietary program information is used.